Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/29/2014
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	Maturing Technology for Major Reductions in Energy, Mass, and Crew Time for Plant Lighting in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	06/09/2009	**End Date:**	06/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is 6/30/2014 per NSSC information (Ed., 5/6/14) NOTE: End date is now 7/31/2014 per NSSC information (Ed., 7/15/13) NOTE: new end date is 7/31/2013 per NSSC, and also is now showing start date of 6/9/2009 (Ed., 6/11/12) NOTE: new end date is 7/31/2012 per NSSC (Ed. 5/3/2011) NOTE: change in Period of Perfomance to 8/1/2009-7/31/2011 (from 6/9/2009-7/8/2011), per NSSC information (Ed., 3/11/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AL99G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To develop and demonstrate proof of concept for two novel technologies that together will greatly advance the potential for growing food crops in space for crew life support at distant space destinations. The first technology is a “smart” LED-based sole-source lighting system for saving considerable energy for many reasons. The second technology is a crop-stand gas-exchange system consisting of a large cuvette, an open gas-exchange system, and computer controls for CO2, temperature, and atmosphere flow rate. Lighting parameters are controlled by a second computer. Using the smart lighting system in conjunction with the crop gas-exchange system will give real-time crop response of hydroponically grown crops to light (PPF, R/B ratio), temperature, and CO2 at all stages of crop production, which will define additional opportunities for significant energy savings during controlled-environment crop production.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project has broad impact for life on Earth as well as in space. On Earth, controlled environment agriculture (CEA) has limitations of profitability, economic viability, and affordability parallel to the issues that hold back the feasibility of using plants for crew life support in space. Interestingly, the single factor that limits the widespread use of CEA on Earth also is that of crop lighting! Supplemental and sole-source lighting using traditional high-intensity discharge (HID) lamps requires major electrical power consumption for waste-heat rejection as well as for lighting per se. The replacement of hot, inefficient HID lamps with solid-state light-emitting diode (LED) technology will benefit CEA and plant research for many reasons: LEDs are small, require much less mass and volume than HIDs, do not require massive ballasts, are extremely long-lived, are relatively cool at photon-emitting surfaces because waste heat is rejected elsewhere, emission wavelengths can be selected to match plant pigments, and can be operated at much lower power levels than HIDs because they can be placed in close proximity to plant tissues. As well, the field of LED technology is progressing at an ever-increasing rate, and electric-to-light conversion efficiencies are increasing at a rapid rate for LEDs. There is great interest in the use of LEDs for CEA on Earth as well as for crew life support in space. In addition, this project addresses the development of “smart” LED lighting systems that automatically sense plant position and size, and subsequently energize LEDs only directly above or in line with plant tissues. When used in conjunction with and supplemental to solar light in greenhouses on Earth, smart LED lighting systems also will monitor the instantaneous contribution of solar light as it tracks across greenhouses daily, and will automatically adjust LED switching and/or dimming/brightening controls in response to the total light environment. The crop gas-exchange system that is part of this project will be used in conjunction with smart LEDs to determine day-to-day crop tolerances, “preferences,” and limits for intensity of light, quality of light, and interaction of light with other important environmental parameters such as CO2 and temperature at all stages of crop development. This capability will greatly accelerate the optimization of environmental conditions needed to produce crops rapidly, efficiently, and affordably, which is just as important for CEA on Earth as it is for closed-loop life support in space.
	NASA Grant NNX09AL99G entitled “Maturing Technology for Major Reductions in Energy, Mass, and Crew Time for Plant Lighting in Space” has tested the hypothesis that leveraging the unique properties of light-emitting diode (LED) technology can significantly reduce the energy requirement for electric plant-growth lighting compared with that required by traditional electrical lighting technology. This research project sought to prove that concept to assist NASA in making future administrative decisions regarding whether or when to re-engage in R & D pertaining to bioregenerative life support for future manned space habitation. The driver for the NASA decision to suspend research on bioregeneration of food was based in part upon experimental data from NASA’s previous Closed Ecological Life-Support System (CELSS) and Advanced Life-support System (ALS) programs. During the CELSS Program in particular, academic research projects sought to define upper limits of photosynthesis, productivity, and yield that candidate food crops for space life support could achieve by eliminating environmental constraints to plant growth and development. Of the five Cardinal factors required for plant growth (i.e., light, temperature, water, nutrients, and atmosphere), light is by far the most expensive in terms of energy utilization. Many realities of habitation on Earth’s moon or Mars will severely constrain the availability of solar light to support growth of higher plants for life support at those space destinations. Therefore, the CELSS and ALS programs focused on hydroponic cultivation of candidate food crops in controlled-environment chambers using electric lighting technologies available for high-intensity lighting at that time (1980s to early 1990s). The dominant electric-lighting technology for plant growth during that era was high-intensity discharge (HID) lighting, and the most popular lamp type was high-pressure sodium (HPS). Intensely hot HPS lamps require considerable spatial separation between lamps and plants (unless chilled water barriers are used), so high power is needed to achieve sufficient photon flux at leaf level to drive photosynthesis and growth. Massive ballasts requiring additional power and which also give off considerable waste heat add to the thermal burden that growth chambers must reject to maintain a favorable thermal environment for plant growth & development. A typical growth chamber lighted with HPS lamps draws 2-3 kW/m2 of growing space for plant lighting and waste-heat rejection. Although CELSS and ALS test crops achieved excellent yield rates of edible biomass, approximately 50 m2 of growing space was calculated to be required per crew member to provide caloric and balanced nutritional needs for space habitation on a continuing basis. Program Managers were comfortable that food crops could support crew diets independent of resupply at distant space destinations, so funding emphasis shifted away from food production to waste management. Privately, however, NASA Program Managers realized that the energy requirements for autonomous food production in space were enormous, and no obvious energy sources in space were identified to support food independence. All bioregenerative life-support research was suspended by NASA in 2006. In the tug of war between energy supply vs. energy use, energy–use efficiency by the lighting system was not previously considered as a place to look for solutions to independent food production in space due to limitations of plant-growth lighting technology available at that time. Energy-use calculations for crop growth generated within the space life-support community were based upon 1980s lighting technology for growth chambers. However, NASA has sponsored low-level research activity with LEDs for plant growth for more than 20 years. This lighting technology has been progressing steadily and is surpassing HID technology in terms of lighting efficacy, with much additional progress predicted. LEDs have multiple advantages over high-voltage gas-discharge lamps: they are solid-state, long-lived, current-driven devices that don’t require high voltage, and can be designed for specific-wavelength emissions that match spectral-absorption profiles of major plant pigments. Most important for the project just completed (NNX09AL99G), LEDs also can be cool to the touch at photon-emitting surfaces, with waste heat being rejected elsewhere. By active heat sinking with flowing air or water, adequate photon emission fluxes for

Task Progress:

plant growth can be achieved from high-output LEDs(>/= 1watt/emitter) without the tissue-scorching dangers of intensely hot HPS lamps. This allows LEDs to be placed close to plant tissues, which in turn requires much less power to deliver adequate photon flux per unit area equivalent to HPS but without high voltage and extreme heat loads.

This unique capability of LEDs previously led to the successful development of vertical, “intracanopy” lighting technology in the PI’s laboratory for tall crops with upper leaves mutually shading lower leaves within closed foliar canopies. In the present project, overhead “close-canopy” lighting was developed for short, rosette-type crops that do not have mutual leaf shading. Special LED arrays were developed in partnership with the Orbital Technologies Corporation (ORBITEC). As for intracanopy lighting with LED “lightsicles” (also custom-developed by ORBITEC for the PI) deployed vertically within the crop stand and LED light engines energized from the bottom up as the crop stand grows in height, close-canopy (CC) lighting allows horizontal LED arrays to be positioned only centimeters above a crop surface. Because overheating of plant tissue by the light source is a non-issue with actively heat-sinked CC LED lighting, spatial separation between arrays and crop surfaces is limited only by the need for light beams from adjacent, different–colored LEDs to overlap before the blended beams fall on the crop surface (e.g., red + blue = purple). Another feature included in this custom-designed LED array was a built-in capability to automatically detect position and size of plants growing in the hydroponic system positioned just below the lamp bank. Leaf lettuce (Lactuca sativa L. cv. Waldmann’s Green) was the rapidly growing model rosette crop used for CC lighting studies in the current project. At pre-programmed intervals, red and blue LEDs driving crop photosynthesis automatically switch off and then green LEDs begin a flash-pattern routine lasting for a few seconds. Green light reflected back up from green leaves is detected by photodiodes distributed between LEDs on the array undersurface. The strength of the flashback pattern indicates where on the hydroponics lid the plants are located and how big they are. An algorithm calculates which LEDs are positioned most directly above plant tissue and subsequently switches back on those red/blue LEDs to resume the photoperiod. ORBITEC gave this smart-lighting system the moniker “High-Efficiency Lighting with Integrated Adaptive Control”, HELIAC for short.

In the present project, two hydroponic crop stands of leaf lettuce were grown in a walk-in controlled-evironment room with its own temperature, humidity, and CO2 controls. HELIAC LED arrays were deployed on height-adjustable mounts above the fixed hydroponic systems. Growth analysis indicated that a lettuce production cycle is characterized by a prolonged lag phase during which seedlings are not very responsive to optimizing conditions of elevated photosynthetic photon flux (PPF) or enriched CO2. Once seedlings develop into the exponential growth phase, they become very sensitive to optimizing growth conditions, but only for a few days. Because lag phase can occupy one third to one half of a lettuce production cycle, strategies were investigated regarding how to save light energy during crop production. When overall PPF was cut back in effort to save energy during lag phase, extreme hypocotyl elongation occurred in the dim light. These overly long (and thin) lower stems became too weak to support seedling growth, causing hypocotyls to break, seedlings to fall over, and die. If PPF was raised during lag, seedlings developed to be shorter, thicker, and stronger with some leaf expansion for future photosynthesis, but little dry mass was gained during lag. Because specific wavelengths of light have different photomorphogenic effects on seedling development, it was decided to keep PPF low during lag to save on overall lighting energy, but to vary red : blue ratios making up that low PPF in effort to keep seedlings strong and viable until they are ready to go into exponential growth. Red:blue ratios of 100:0, 95:5, 90:10, or 80:20 were tested in preliminary experiments using initial lag-phase low PPFs of 75 or 100 µmol/m2/s. Hypocotyl elongation decreased with increasing proportion of blue light, but so also did expansion of the first set of true leaves, which limited effective photosynthetic surface area as seedlings came out of lag phase. A compromise R:B ratio of 95:5 was found through experimentation that kept seedlings from excessive “stretching” during lag but did not inhibit leaf expansion. The same R/B ratio worked for sole-source LED lighting both for exponential as well as for lag phase and was subsequently adopted for the duration of the study.

Using a 95:5 R/B ratio, it was determined for the remaining environmental conditions of these experiments that the lag phase of a lettuce-production cycle lasts for 15 days before plants go into rapid exponential growth, at which time PPF was raised to 350 µmol/m2/s and CO2 enriched from normal ambient (about 400 µmol/mol) to 1000 µmol/mol. Plants were harvested on day 21 after germination because beyond that time the second inflection point of the growth curve occurred, after which growth started to curve over toward the plateau phase of a sigmoid growth curve and thus becomes increasingly less responsive to optimizing environments once again. Thus, for comparative purposes between LED lighting treatments, it was decided to compare cumulative lighting-system efficacies from germination until right before the onset of plateau growth for this LED system. Using CC LED lighting with 95:5 R/B lighting, power utilized for lighting per unit growth area was 202 W/m2 compared to a power density of 2100 W/m2 for HPS lighting in NASA’s Biomass Production Chamber (BPS). Thus, CC LED lighting consumes an order-of-magnitude less instantaneous power per unit growth area than does dated HPS plant-lighting technology. If energy utilized for lighting is expressed per unit dry plant biomass formed, then the energy cost of CC R + B LED lighting was 1.16 kWh/g dry biomass in our study compared to 4.7 kWh/g for ‘Waldmann’s Green’ lettuce grown in NASA’s BPC, which is about a 75% improvement in energy-to-biomass conversion efficiency for LEDs. When targeted CC LED lighting was compared with total coverage CC LED lighting, energy cost per unit dry plant biomass formed decreased to 0.62 kWh/g, a further improvement of about 50% in energy-use efficiency without photons falling on empty spaces between plants. The combination of targeted and CC LED lighting amounts to an 8-fold improvement in energy-use efficiency compared to the dated HPS lighting technology used in the BPS. It is a great start on a first order-of-magnitude energy reduction for bioregenerative food production in space. Thus, using one-tenth the power density for crop lighting, the CC/targeted-lighting strategy of the present investigation improved light-to-biomass-conversion efficiency 8-fold over that of dated HPS lighting technology.

The efficiency of electrical energy conversion to light by LEDs available at the time that HELIAC was constructed for this study was 29% for the reds and 41% for the blues. Several years later, the efficiency of red LEDs had improved to 38% while that of blue LEDs improved to 49%. Improvements such as these are predicted to continue into the forseeable future. Thus, the prospects for energetically favorable food production in space based upon 2006 LED lighting technology will only improve over time with ongoing advances in solid-state lighting technology.

In addition to performing this proof of concept, the other emphasis of NASA Grant NNX09AL99G has been to develop a research tool that will enable strategic management of the plant-growth environment so that costly resources are not wasted during crop production but plants also are not deprived of growth-limiting resources when they need them, but only when they need them. The tool is intended to continuously monitor real-time plant response to several major environmental inputs over the entire course of a crop-production cycle, and to manage that total growth environment in ways that minimize energy inputs without compromising crop yield.

To do that, we have constructed a crop-stand gas-exchange cuvette for relatively small-stature crops that is integrated

with an open gas-exchange system, a recirculating deep-batch hydroponics system, and a HELIAC plant-lighting unit. The cuvette is sized to interface simultaneously with the HELIAC as well as a rectangular hydroponic unit whose lid dimensions accommodate up to 49 leaf-lettuce plants widely spaced as young seedlings but which grow together to form a closed foliar canopy over the course of a 25-30 day crop-production cycle. The cuvette has an expandable bellows-type of wall construction that accommodates increases in crop height during growth while keeping constant vertical separation distance between the bottom of the lighting array and the top of the crop. The entire assembly is located within a walk-in growth room having its own thermal control. The entire assembly is tightly clamped together to prevent leaks (especially inward!). The rectangular cuvette has an inlet port at one end and an outlet port at the opposite end. An assemblage of blower, compressed-gas cylinders, mass-flow-control valves, gas-flow plumbing, and a control computer located exterior to the growth chamber deliver air with a programmed concentration of CO2 at a precise flow rate to the inlet port of the cuvette. Fans located within the cuvette mix the inlet air thoroughly across the width and along the length of the cuvette. Plants growing in the cuvette remove CO2 from the air (in the light) as it flows turbulently toward the outlet port on the opposite side of the cuvette space. Air exiting the cuvette contains less CO2 than air entering the inlet port by an amount reflecting light intensity, CO2 concentration, plant size (leaf area), and flow rate of air within the cuvette. Just prior to the inlet, the gas stream tees off and a small slipstream bypasses the cuvette. The outlet and bypass gas streams flow out of the growth chamber to a gas-mixing board, from which small aliquots of both sample (outlet) and reference (bypass) streams flow at controlled, slow rates through the cells of a differential infrared gas analyzer (d-IRGA). The delta CO2 detected by the d-IRGA is logged on the control computer, which multiplies delta CO2 by airflow rate entering the cuvette and divides it by the total hydroponic growth area and does unit conversions to express photosynthetic rate (Pn) as µmol/m2/day. Standard CO2 span gases and an absolute IRGA are used to periodically calibrate the system for accuracy and correct for electronic drift.

The goal is to monitor crop Pn over time from as early as detection of photosynthesis is possible for a seedling stand to harvest of a mature crop with a continuous, closed canopy. Flow rate of air through the system will be increased proportionally to crop growth (leaf area) over time to maintain delta CO2 as small and constant as possible. Daily crop Pn measurements will reflect the sigmoid growth curve of the crop, without destructive sampling of plants within the cuvette for biomass measurements. Leaf lettuce will be used as the initial test species whose growth rate closely tracks its photosynthetic rate, which is highly responsive to light intensity, CO2 concentration, and temperature. These three major environmental parameters will be manipulated in ways that induce significant crop-stand response to one or more factors at key stages of crop development (e.g., during lag phase, exponential growth, plateau phase). Water status and mineral nutrition will be carefully managed (in the hydroponic reservoir) so that neither factor becomes limiting during the course of a crop-production cycle. Thus, photosynthesis will be used as a real-time (and rapidly responding) metric of crop growth. By changing set points for CO2 concentration at the control computer, e.g., it will be possible to determine when the crop is coming out of lag phase and is becoming responsive to optimizing environments, and to do that without destructive sampling for growth analysis (for example, dry weight determination). By briefly increasing airflow rate through the cuvette after an up or down tweak in CO2 or PPF, it will be possible to quickly re-equilibrate steady-state CO2 levels for determination of a new Pn, and then return airflow to its previous or new level. For light, the HELIAC control computer will accommodate immediate increments of increase or decrease in photosynthetic photon flux (PPF), and the cuvette system will detect any changes in Pn to a new steady-state rate within minutes. This will be very valuable for determining, for example, when to back off on light intensity and CO2 enrichment when crop development is moving from exponential to plateau phases of growth, and by how much. As well, it will be possible to change red:blue ratios of LED lighting immediately, to detect initial crop response within seconds (i.e., delta CO2 up or down on the computer screen), and see the new steady-state rate of Pn reported within minutes. For example, under high CO2, which tends to close leaf stomates (pores), if blue light is increased, which tends to open stomates, the degree of Pn stimulation might be extended to higher-than-normal maximum CO2 levels (for high Pn) by increasing the proportion of blue light.

Due to the complex multiple components of the gas-exchange / cuvette / hydroponics / lighting system, plus the restrictions of a modest budget, it has been necessary to design, fabricate, build, test, troubleshoot, and refurbish components of the integrated plant-monitoring system using local resources and student projects rather than the services of a professional engineering corporation, which we wouldn't have been able to afford. Other than HELIAC, which was developed in collaboration with ORBITEC, a (then) small business with strong interests in the space program, the rest of the crop-monitoring system has been developed locally. A design-build local engineer developed cuvette as a side job, and has been through several iterations of retrofit and refurbishment as our integration tests indicated. Purdue University Engineering and Engineering Technology students, both undergraduate as well as graduate, have contributed to development of the open air-flow system, to programming for control of CO2 concentration and air flow, data logging, calculating, and reporting of gas-exchange parameters, and to development of a uniform airflow and thermal-control environment within the cuvette per se. Life science undergraduates collaborate with engineering students to develop protocols for growing test crops hydroponically, to measure gas exchange, manage data, and perform routine system maintenance. At the end of the extended project all development tasks still were not completed because of limited student time, turnover, graduations, etc., but the effort to develop a fully integrated, operational crop-growth / multi-plant gas-exchange system continues using other funding sources. It is planned for a leak-free, thermally stabilized open gas-exchange system to be completed by the end of 2014, and for proof-of-concept crop-growth / gas-exchange studies to be conducted during the spring semester of 2015. That study will establish interactive dose-response curves for Pn vs. PPF at several different CO2 levels at each important phase of a crop-production cycle, and for those experiments to be replicated statistically. A key objective of that study will be to identify precisely when during the time course of a crop-production cycle that optimizing environments are needed, what the minimum inputs are, and when they no longer are needed. The outcome of that study is expected to identify additional times and ways in which significant resources (e.g., CO2) and energy (e.g., light, electricity) can be saved during crop production, and that is expected to additionally reduce costs of growing food in space. The current plan is to complete the proof-of-concept study by May, 2015, write up and internally review a manuscript for publication, and submit an article for publication to a peer-reviewed journal by fall, 2015. An army of past and present engineering, engineering technology, and life science students will be co-authors on that proof-of-concept article that will come out of this technology-development effort enabled by NASA grant NNX09AL99G. This NASA grant will be acknowledged as the main funding source supporting the new research technology proof-of-concept work. Once published and operational, this tool will be proposed for data gathering and further upgrade in future grant proposals. It is anticipated to provide incentive for future fund raising because of it's powerful analytical capabilities.

Bibliography Type: Description: (Last Updated: 04/16/2020)

Abstracts for Journals and Proceedings	Poulet LBN, Massa GD, Morrow RC, Bourget CM, Wheeler RM, Mitchell CA. "Significant reduction in energy for plant-growth lighting in space using targeted LED lighting and spectral manipulation." To be presented at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. Program and abstracts, in press. , Oct-2014
Articles in Peer-reviewed Journals	Poulet L, Massa GD, Morrow RC, Bourget CM, Wheeler RM, Mitchell CA. "Significant reduction in energy for plant-growth lighting in space using targeted LED lighting and spectral manipulation." Life Sciences in Space Research. 2014 Jul;2:43-53. http://dx.doi.org/10.1016/j.lssr.2014.06.002 , Jul-2014